ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

AGRICULTURAL TECHNOLOGY

PAPER 1

MARKS: 100 **TIME: 2 hours**

Topics: safety, materials and tools, fencing and farm structures, energy, and water supply.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions. Show ALL calculations, including formulae and units.

QUESTION 1

1.1 Choose the correct answer. Write only the letter (A–D).

1.1.1 Which protective equipment protects the eyes during arc welding?

A Ear plugs
B A welding helmet with a dark filter lens
C A dust mask
D Leather gloves

1.1.2 Galvanised steel is coated with … to prevent rust.

A paint
B zinc
C copper
D tin

1.1.3 The tool used to tighten fencing wire is a …

A wire strainer.
B hacksaw.
C spirit level.
D chisel.

1.1.4 Which energy source is renewable?

A Coal
B Diesel
C Solar
D Natural gas

1.1.5 The unit of electrical power is the …

A volt.
B ampere.
C watt.
D ohm.

(5 × 2)
(10)

1.2 Give ONE term for each description.

1.2.1 A mixture of two or more metals, such as brass or bronze

1.2.2 The strong post at the end of a fence line that holds the strain of the wires

1.2.3 A deep, narrow hole drilled to reach underground water

1.2.4 Joining two metal pieces by melting them together

1.2.5 A fence that gives animals a short, safe electric shock to keep them in a camp (5 × 2)
(10)

[20]

QUESTION 2: SAFETY, MATERIALS AND TOOLS

2.1 Give FIVE general safety rules for working in a farm workshop. (5)

2.2 Distinguish between ferrous and non-ferrous metals. Give ONE example of each. (4)

2.3 Explain why steel rusts, and give TWO ways to prevent rust. (4)

2.4 Name TWO hand tools used for cutting metal and ONE tool used for measuring. (3)

2.5 Give FOUR guidelines for looking after (maintaining) hand tools. (4)

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QUESTION 3: FENCING AND FARM STRUCTURES

3.1 Give THREE reasons why fences are important on a farm. (3)

3.2 A farmer wants to fence a rectangular camp of 400 m by 250 m with 5 strands of wire.

3.2.1 Calculate the perimeter of the camp. (2)

3.2.2 Calculate the total length of wire needed. (2)

3.2.3 Droppers are placed every 5 m. Calculate how many droppers are needed. (2)

3.2.4 Wire costs R1 250 per 500 m roll. Calculate the cost of the wire. (3)

3.3 Describe the steps in erecting a straining post (end post). (4)

3.4 Name FOUR materials used to build farm structures such as sheds and kraals. (4)

[20]

QUESTION 4: ENERGY ON THE FARM

4.1 Name FOUR sources of energy that a farm can use, and say whether each is renewable or non-renewable. (8)

4.2 An electric water pump has a power rating of 750 W and runs for 6 hours per day. Electricity costs R2,50 per kWh.

4.2.1 Calculate the energy used per day in kWh. (3)

4.2.2 Calculate the cost of running the pump for 30 days. (3)

4.3 Give THREE advantages of using solar panels on a farm. (3)

4.4 Give ONE safety rule when working with electricity on a farm. (3)

[20]

QUESTION 5: WATER SUPPLY

5.1 Name FOUR sources of water for a farm. (4)

5.2 Name TWO types of water pumps used on farms. (2)

5.3 A cylindrical water tank has a diameter of 2 m and a height of 1,5 m. Calculate its volume in litres. Use $\pi = 3{,}14$ and $1\ m^3 = 1\ 000\ \ell$. (5)

5.4 Give FOUR ways to save water on a farm. (4)

5.5 Explain why water for animals should be tested regularly. (5)

[20]

TOTAL: 100

END OF QUESTION PAPER